Lab_07 Model Evaluation

ITAI 1371

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Reflection

Before this lab, I thought accuracy was more important than recall. Now I realize precision and recall measure different mistakes: precision checks false alarms, and recall checks misses. I have to think about which mistake is worse to decide which one matters more for what I am trying to achieve. This lab also helped me understand the importance of cross-validation. Upon the single split, the model scored 80.4%, which seemed “good enough.” Cross-validation broke the data into 5 folds, and each round tested a different 20%. It averaged 78.68%, which showed the single split got “lucky”.

In my knowledge check answers, I used my bank card and smoke detector examples, which helped me realize that precision is not automatically preferred over recall. Both are valuable, and each gives a different view of the model’s mistakes. With the bank card, precision mattered more because a false fraudulent flag on the card could cause embarrassment. So being right when it flags fraud matters most in this case. However, with the smoke detector example, recall mattered more because you’d rather have false alarms than missing small real fires.

The notebook file was broken. It had lost its line breaks, so all the code was mashed onto one line. I split each command so it could be read properly. When I ran cell 1, I received 2 errors, first ChainedAssignmentError, then a ValueError. The model refused to train because “Age” still had blanks (NaN). My newer pandas (version 3) ignored inplace=True, so the blanks never got filled. On my first try, I removed inplace=True, but the same error came back because the filled-in column wasn't saved anywhere. I added df[‘Age’] = at the start of the line so the filled-in values were saved back to the column.. After that, all 3 tasks ran successfully, the confusion matrix, the classification report, and cross-validation.

Next time, I'll check precision and recall instead of only looking at accuracy, as well use cross-validation for a more dependable score.

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Before this lab, I mostly looked at accuracy to decide whether a machine learning model was performing well. My Titanic model had about 80% accuracy, so at first I thought that was enough to say the model was good. After completing this lab, I learned that accuracy alone does not explain everything about a model's performance. The confusion matrix helped me see exactly where the model was correct and where it made mistakes.

My confusion matrix showed that the model correctly predicted 90 passengers who did not survive and 54 passengers who survived. It also made 35 mistakes, including 15 false positives and 20 false negatives. This helped me understand that two models could have similar accuracy but make very different kinds of mistakes. In this case, the model missed more actual survivors than it falsely predicted survivors. That information would not be clear if I only looked at the overall accuracy.

Another important part of this lab was learning the difference between precision and recall. At first, these two ideas were a little difficult for me because they both measure model performance but in different ways. Working with my own results made them easier to understand. The recall for survivors was 0.73, which means the model correctly identified about 73% of the passengers who actually survived. It missed about 27% of them. I learned that precision tells me how often the model is correct when it predicts a positive case, while recall tells me how many of the real positive cases the model successfully finds.

The classification report also helped me see that the model performed better for passengers who died than for passengers who survived. For the "Died" class, the F1-score was 0.84, while for the "Survived" class it was 0.76. This showed me why looking at each class separately is important, especially when the classes are not balanced.

Cross-validation was another important lesson. My single train-test split gave about 80.4% accuracy, but the 5-fold cross-validation average was about 78.68%. The scores ranged from about 77.5% to 80.3%. This showed me that one train-test split can give a result that is slightly better or worse depending on which data points are selected. The small standard deviation of 0.0094 showed that the model was still fairly stable across the different folds.

This lab also connects directly to my stroke prediction project. My stroke dataset has very few positive stroke cases compared with no-stroke cases. Because of this, I now understand that a model could have high accuracy and still fail to identify the people who actually had a stroke. For that type of problem, I need to pay close attention to recall, precision, F1-score, the confusion matrix, and cross-validation instead of depending only on accuracy.

Overall, this lab changed the way I think about model evaluation. I no longer want to ask only, "What is the accuracy?" I also want to know what kinds of mistakes the model makes, which group it predicts better, and whether its performance stays consistent when it is tested on different parts of the data.

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derrick gibbs
Here is my reflection journal.

This lab helped me understand why accuracy alone does not tell the whole story about a machine learning model. Before working through the results, it was easy to look at one score and decide whether a model worked well. Using the Titanic data made me think more carefully about the different kinds of mistakes a model can make. The goal was to predict whether a passenger survived, but being correct most of the time does not mean every type of prediction was equally reliable. The confusion matrix made this clearer for me. The model correctly predicted 90 passengers who did not survive and 54 who survived. It also predicted that 15 passengers survived when they did not, and that 20 did not survive when they did. Seeing those numbers helped me understand the difference between correct predictions and mistakes. It also showed that the model missed more actual survivors than it incorrectly predicted as survivors. I understood the terms true positive, true negative, false positive, and false negative better when I connected them to people in the dataset instead of only memorizing their definitions. The classification report gave me another way to look at performance. The model had about 80% accuracy, but its recall for survivors was 73%, compared with 86% for passengers who did not survive. Its F1-score was also lower for survivors, at 0.76 compared with 0.84 for passengers who did not survive. This showed me why precision, recall, and F1-score matter. Depending on the problem, one type of mistake could be more important than another. For example, with a medical screening test, missing someone who may be sick could be a serious problem, even if the overall accuracy looks good. Cross-validation was another important part of this lab. The five-fold cross-validation average was 78.68%, which was close to the 80% accuracy from the single test split. The scores also varied very little, with a standard deviation of 0.0094. I learned that testing a model on different parts of the data can give a better picture of how consistently it performs. At the same time, close scores do not mean a model is perfect. They only show

that its results were consistent for the data used in this lab. The main challenge for me was understanding what each evaluation measure adds beyond accuracy. Comparing the confusion matrix with the classification report helped the numbers make sense. This also connected to earlier work with training and testing models. Training a model is only ne part of machine learning; checking its predictions is just as important. If I continued this project, I would want to explore whether other passenger information could improve the predictions. I would also compare results carefully instead of choosing a model based on accuracy alone.